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19 January 1977
SUPERSEDING
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26 November 1968

MILITARY SPECIFICATION

MULTIPLIER: HIGH VOLTAGE, DIRECT CURRENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of solid state voltage multipliers which are a part of the Night Vision Image Intensifier Assemblies.

1.2 Classification. The voltage multipliers covered by this specification shall be classified as follows:

Type I	The type I voltage multiplier shall be one piece construction, for use on types MX-7854/UV and MX-8501/UV (25MM) Image Intensifier Assemblies (See Figure 1).
Type II	The type II voltage multiplier shall be one piece construction, for use on the type MX-7856/UV (40MM) Image Intensifier Assembly (See Figure 2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-636	- Box, Shipping Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Cdr, USA Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-P-26514	- Polyurethane Foam, Rigid, or Flexible for Packaging

STANDARDS

FEDERAL

FED-STD-356	- Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-781	- Reliability Testing, Exponential Distribution

DRAWINGS

ELECTRONICS COMMAND

SC-D-611632	- Multiplier: High Voltage
SC-D-611738	- Multiplier: High Voltage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The high voltage direct current (Vdc) multiplier, hereinafter called the "multiplier" shall be a solid state electronic device to convert a sinusoidal alternating current (ac) input voltage of 2700 volts, ±100 volts peak to peak at a frequency of 1500 hertz (Hz), ±100 Hz to multiple high dc output voltages. The multiplier shall be wrap-around construction conforming to Figures 1 and 2 and as specified herein. Unless otherwise specified, all performance requirements apply to both types of voltage multipliers (see 1.2).

3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3 and 6.2.

3.3 Construction. The multiplier shall be constructed in accordance with the requirements of this specification and SC-D-611738and SC-D-611632.

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3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Dimensions and weight.

3.5.1 Dimensions. Dimensions shall be as shown on the applicable figure (see Figure 1 or 2) and as specified herein.

3.5.2 Weight. The type I multiplier shall not exceed 4 ounces total weight. The type II multiplier shall not exceed 5 ounces total weight.

3.6 Operating and environmental.

3.6.1 Operational requirements.

3.6.1.1 Output voltage no-load. The output voltage, no-load, (see 6.3.2) shall vary not more than 7 percent from the actual voltage readings (see 3.6.4) taken at room temperature when operating at -54°C.

3.6.1.2 Output voltage V_3. The output voltage, V_3, shall not exceed 53,000 volts while operating at +52°C.

3.6.1.3 Multiplier charging current. The multiplier changing current shall not exceed 150 microamperes while operating at +52°C measured from the zero reference line to the peak of the charging spike on an oscilloscope display.

3.6.2 Multiplier burn-in. Each multiplier shall be operated continuously for 100 hours under no-load condition in a grounded tube simulator prior to any quality assurance inspections.

3.6.3 Continuity. The multiplier shall exhibit continuity between output terminal GRD-1 and V_3.

3.6.4 D.C. output voltages. With the multiplier mounted in a tube simulator and leads terminated with corona surpressing balls (see Figure 4), the multiplier shall meet the following requirements when operated with an input potential of 2785 volts, ±15 volts (peak to peak) at a frequency of 1500 Hz, ±100 Hz:

a. With R, S_1 and S_2 open circuited, output voltages shall be:

GRD-1 to V_1A = 12,300, ±500 Vdc

GRD-1 to V_1 = 14,600, ±500 Vdc

GRD-1 to V_2 = 28,100, ±750 Vdc

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GRD-1 to V_3	= 45,400, ±2,000 Vdc
V_2 to V_3	= 17,600, +1,000, - 2,000 Vdc
Switching from GRD-1 to GRD-2 will drop each voltage by	2,300, ±300 Vdc

b. With S_1 closed, R and S_2 open circuited, GRD-1 to V_1 and GRD-1 to V_1A shall not vary more than 150 Vdc from the value measured in (a) above.

c. With R=40 G-ohms, S_1 and S_2 open circuited, GRD-1 to V_3 shall not vary more than 1,000 Vdc from the value measured in (a) above.

d. With S_2 closed, S_1 and R open circuited, GRD-1 to V_2 shall not vary more than 600 Vdc from the value measured in (a) above.

e. With S_1 and S_2 open, vary R from 1 megohm to 500 G-ohm. Measure $V_3 - V_2$ and I_3 as R is varied to produce $V_3 - V_2$ voltage steps of 1,000, ±200 Vdc. When $V_3 - V_2$ is plotted versus I_3, the plotted curve shall fall below the curve shown in figure 5.

3.6.5 Resistance third stage, limiting. The resistance value of the third stage limiting resistor shall not vary more than 20 percent from the suppliers design center when tested in accordance with 4.6.5.

3.6.6 Ground probe resistance. With the multiplier input potential adjusted to produce 30,000 Vdc between GRD-1 to V_3, there shall be no evidence of arcing, corona or multiplier body breakdown when a ground probe (see Figure 7) is scanned over all surfaces of the multiplier.

3.6.7 Operational safety. The multiplier shall not be damaged when the output terminals are grounded for 5 seconds and the multiplier is operating with an input potential of 2800 volts, ±15 volts (peak to peak) at a frequency of 1500 Hz, ±100 Hz.

3.6.8 Input voltage waveform. With the multiplier connected to the test circuit as shown in Figure 8 and while operating with an input potential of 2800 volts, ±15 volts (peak to peak) and a frequency of 1500 Hz, ±100 Hz, the instantaneous input voltage waveform shall be observed. The waveform shall appear sinusoidal with the exception of two charging spikes per cycle approximately 180° apart. There shall be no sharp breaks in the curve or evidence of corona, breakdown or arcing which appears as nonuniform spikes or intermittent distrubances in the waveform (see Figure 3).

3.6.9 Leads. All multiplier lead wires shall be stranded #22 or #24 gauge heavy walled polyvinyl chloride insulated wire. Nominal wall thickness of the insulation shall be a minimum of 0.029 inch. The wire shall be pretinned. V_2 and V_3 output leads shall be covered throughout their exposed lengths with

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polyolefin or polyvinyl chloride heat shrinkable tubing. This tubing shall be installed prior to encapsulation of the multiplier. The insulated portion of the leads shall be potted into the body of the multiplier. The leads shall not fracture or show evidence of deterioration of insulation characteristics after being subjected to 12 each, 60° bends as shown in Figure 9.

3.6.10 Mean time between failure. The mutlipliers shall exhibit a mean time between failure (MTBF) of 4,000 hours.

3.6.11 Storage temperature. The multiplier shall not be damaged (see 6.3.1) and shall meet specified operational requirements after being subjected to storage at a temperature of +68°C, ±3°C and minimum relative humidity of 90 percent for 24 hours and at a temperature of -54°C for 24 hours.

3.6.12 Thermal shock. The multiplier shall not be damaged and shall meet the operational requirements specified below when tested as specified in 4.6.12.

3.6.13 Altitude. There shall be no evidence of arcing, corona or any other intermittent or continuous failure which appears as nonuniform spikes present in the E_{p-p} waveform (see Figure 3) when the multiplier is operated with 3,000 volts (peak to peak) input voltage at an atmospheric pressure equal to a 10,000 foot altitude and an ambient temperature of -54°C.

3.6.14 Shock. The multiplier shall not be damaged, shall show no evidence of arcing, corona or any other intermittent or continuous failure as evidenced by nonuniform spikes present on the E_{p-p} waveform and shall meet all operational requirements while being subjected to six shock impacts in each of three mutually perpendicular planes with the multiplier operating at 2800 volts, ±15 volts (peak to peak). The shock impacts shall be nominal half sine wave pulses having a minimum of 100 g's (see 6.3.3) at peak amplitude. The duration of each shock pulse shall be 6 milliseconds, ±2 milliseconds measured between the 10 percent values of the peak amplitude. The energy under the curve shall be not less than 0.25 g-second and the after oscillations shall be not greater than 15 percent of the peak amplitude of the nominal half sine wave pulse.

3.6.15 Vibration (1), (Type I multiplier). The multiplier shall not be damaged and shall meet all requirements of 3.6.4 after being rigidly mounted, singly or in groups, and vibrated with simple harmonic motion in three mutually perpendicular planes at a constant frequency of 30 Hz with an amplitude of 0.125 inch total excursion for 5 minutes in each plane. There shall be no voltage applied to the multiplier during the vibration.

3.6.16 Vibration (2), (Type I and II multipliers). The multiplier shall not be damaged and shall meet all requirements of 3.6.4 after being rigidly mounted, singly or in groups, and vibrated with simple harmonic motion in three mutually perpendicular planes over a frequency range of 10 to 55 Hz with an amplitude of 0.10 inch total excursion for 10 minutes in each plane. There shall be no voltage applied to the multiplier during the vibration.

3.6.17 Marking. Marking shall be as shown in drawing SC-D-611738 in accordance with MIL-STD-130. Marking color shall be in high contrast to the multiplier color. EXAMPLE: Black letters and numerals on white background.

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3.6.18 Workmanship. The multiplier shall be free of burrs, sharp edges, potting voids, defects, scratches, and other mars. It shall also be free of grease, oil, dust, solder, flux and any other conductive film. The processes of fabrication and non-destructive testing shall not deteriorate the characteristics of multiplier performance, material, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.5).

c. Inspection of preparation for delivery (see 4.7).

4.3 First article inspection. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 Inspection. Each multiplier shall be examined in accordance with Table I. One or more defects shall be cause for rejection of that multiplier.

TABLE I Inspection.

Defects	Requirement paragraph
Major	
101. Dimensions not as specified	3.5.1
102. Parts and materials not as specified	3.4
103. Marking not in accordance with MIL-M-13231	3.6.17
104. Workmanship not as specified	3.6.18
105. Weight not as specified	3.5.2

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4.3.2 Tests. Upon successful completion of the inspections specified in 4.3.1, the first article multiplier shall be subjected to the inspections in Table II. Inspections may be conducted in any sequence. Failure of any test shall be cause for rejection.

4.4 Test conditions. Tests shall be conducted in the order listed in the tables. Unless otherwise specified in the test procedures, test conditions shall be as follows:

a. Multiplier shall be mounted in the tube simulator as shown in Figure 4 or an equivalent test fixture.
b. The oscillator used to supply the input voltage shall supply 2800 volts at 1500 Hz under a 10 milliampere load. The oscillator shall deactivate when the load exceeds 10 milliamperes.
c. Output voltages are V_1, V_2, V_3 of Figure 4.
d. Chambers used for temperature tests shall maintain the specified temperature within $\pm 2°C$.
e. Each multiplier subject to quality assurance inspection shall be operated with an input of 2800 V, ± 15 at a frequency of 1500 Hz, ± 100 Hz for 100 hours in accordance with 3.6.2 (see 4.6.2).
f. All tests shall be conducted at room temperature.

TABLE II First article inspection.

Inspection	Requirement paragraph	Test paragraph
Multiplier burn-in	3.6.2	4.6.2
Continuity	3.6.3	4.6.3
D.C. output voltages	3.6.4	4.6.4
Resistance (third stage, limiting)	3.6.5	4.6.5
Ground probe resistance	3.6.6	4.6.6
Input voltage waveform	3.6.8	4.6.8
Operational safety	3.6.7	4.6.7
Multiplier input capacitance	3.6.13	4.6.1
Leads	3.6.9	4.6.9
Thermal shock	3.6.12	4.6.12
Altitude	3.6.13	4.6.13
Shock	3.6.14	4.6.14
Vibration (1)	3.6.15	4.6.15
Vibration (2)	3.6.16	4.6.16
Storage temperature	3.6.11	4.6.11
Mean time between failure	3.6.10	4.6.10

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified in the contract or purchase order, for purposes of inspection a lot shall be as defined in MIL-STD-105.

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4.5.2 Sampling. Samples shall be selected in accordance with MIL-STD-105. AQL and inspection level shall be as specified in Tables IV and V.

4.5.3 Examination. The multipliers shall be examined in accordance with Table I. AQL and inspection level shall be as specified in Tables IV and V.

4.5.4 Tests.

4.5.4.1 Group A Inspection. Each multiplier shall be subjected to the tests in Table III. Failure of any test shall be cause for rejection.

TABLE III Group A Inspection.

Inspection	Requirement paragraph	Test paragraph
Multiplier burn-in	3.6.2	4.6.2
Continuity	3.6.3	4.6.3
D.C. output voltages	3.6.4	4.6.4
Resistance (third stage, limiting)	3.6.5	4.6.5
Ground probe resistance	3.6.6	4.6.6
Input voltage waveform	3.6.8	4.6.8

4.5.4.2 Group B Inspection. Samples selected in accordance with 4.5.2 and which have passed the tests specified in 4.5.4 shall be subjected to the tests in Table IV. AQL shall be 1.0 percent defective, inspection level S-3 combined. Failure of any test shall be cause for rejection.

TABLE IV Group B Inspection.

Inspection	Requirement paragraph	Test paragraph
Operational safety	3.6.7	4.6.7
Multiplier input capacitance	3.6.13	4.6.1
Leads	3.6	4.6.9
Thermal shock	3.6.12	4.6:12
Altitude	3.6.13	4.6.13
Shock	3.6.14	4.6.14
Vibration (1)	3.6.15	4.6.15
Vibration (2)	3.6.16	4.6.16

4.5.4.3 Group C Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and passed the tests of 4.5.4.1 and 4.5.4.2 inspections. Sample units shall be selected in accordance with 4.5.2 and 4.5.5.

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TABLE V Group C Inspection.

Inspection	Requirement paragraph	Test paragraph	AQL Percent defective	Inspection level
Storage temperature	3.6.11	4.6.11	1	S-3
Mean time between failure	3.6.10	4.6.10	-	-

4.5.5 Special sampling for inspection. Sample multipliers shall be selected at random for each inspection in Table V each 30 days. The sample size shall be determined by the AQL and inspection level indicated in Table V. The lot size shall be the number of multipliers accepted in the preceeding 30 days.

4.5.6 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2).

4.6 Test methods. Tests required to determine compliance with the requirements of this specification shall be conducted in accordance with the test procedures specified herein.

4.6.1 Multiplier input capacitance. Mount the multiplier in the tube simulator as shown in Figure 4 and connect as shown in Figure 8. Operate the multiplier with an input of 2800 volts, ±15 volts (peak to peak) at a frequency of 1500 Hz, ±100 Hz. Measure the peak to peak input current and compute the capacitance by

$$C = \frac{I_{p-p}}{2 \quad fE_{p-p}}$$

WHERE: I_{p-p} is the input current
E_{p-p} is the input voltage
f = 1500 Hz

Failure to meet the requirements of 3.6.1.3 shall constitute failure of this test.

4.6.2 Multiplier burn-in. Each multiplier shall be subjected to a 100 hour burn-in period per test conditions (see 4.4).

4.6.3 Continuity. Connect the multiplier to the circuit as shown in Figure 10. Failure of the lamp to light and remain on indicates a break in diode continuity and constitutes failure of this test.

4.6.4 D.C output voltages. Mount the multiplier in the tube simulator (see Figure 4) and terminate the leads with corona surpressing balls. Connect the multiplier as shown in the test circuit in Figure 8. Failure to meet the requirements of 3.6.4 shall constitute failure of this test.

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4.6.5 Resistance (third stage, limiting). Connect the multiplier as shown in Figure 6. Compute R using the formula:

$$R \text{ multiplier} = \frac{E900 - E600}{e900 - e600} \times 5 \times 10^5$$

Failure to meet the requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Ground probe resistance. While operating with a potential of 30 kV on V_3, rub the ground probe over all surfaces of the multiplier. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Operational safety. Operate the multiplier with an input potential of 2800 volts, ±15 volts, (peak to peak) at a frequency of 1500 Hz, ±100 Hz. Ground each output terminal for 5 seconds. Any damage to the multiplier shall constitute failure of this test.

4.6.8 Input voltage waveform. Connect the multiplier as shown in Figure 8. Operate the multiplier with an input of 2800 volts, ±15 volts (peak to peak) and at a frequency of 1500 Hz, ±100 Hz and observe the instantaneous input voltage waveform. Failure to meet the requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Leads. Bend the leads on the multiplier as shown in Figure 9. Failure to meet the requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Storage temperature. Place the multiplier in the temperature chamber at +68°C and a minimum relative humidity of 90 percent for 24 hours. Lower the temperature to -54°C and allow the multiplier to remain for 24 hours. After removal from the chamber examine the multiplier. Evidence of damage or failure to meet the requirements of 3.6.11 shall constitute failure of this test.

4.6.11 Thermal shock. Adjust the chamber temperature to -54°C. Place multiplier in the temperature chamber and allow the mutliplier temperature to stabilize for 1 hour. Operate the multiplier at -54°C for 30 minutes. Remove the multiplier and allow it to reach room temperature. Adjust the chamber temperature to +52°C. Place the multiplier in the chamber. Allow the multiplier temperature to stabilize for 1 hour. Operate the multiplier at +52°C for 1 hour. Remove the multiplier and allow it to reach room temperature. Operate the multiplier. Failure of the multiplier to operate in accordance with 3.6.12 shall constitute failure of this test.

4.6.12 Altitude. Place the multiplier in a test chamger and reduce the chamber temperature to -54°C and adjust the atmospheric pressure to an equivalent pressure of 10,000 feet above sea level. Allow the multiplier to stablize for 1 hour. Operate the multiplier with an input of 3000 volts (peak to peak) for 10 minutes. Any evidence of arcing, corona, or any other intermittent or continuous failure which appears as nonuniform spikes present in the E_{p-p} waveform (see Figure 3) shall constitute failure of this test.

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4.6.13 Shock. Secure the multiplier in the test fixture and operate with 2800 V peak to peak input voltage. While operating, shock the multiplier with six impacts in each of three mutually perpendicular planes. The shock impacts shall be nominal half sine wave pulses having a minumum of 100 g's at peak amplitude. The duration of each shock pulse shall be 6 milliseconds, ±2 milliseconds measured between the 10 percent values of the peak amplitude. The energy under the curve shall be not less than 0.24 g-seconds and after oscillations shall be not greater than 15 percent of the peak amplitude of the nominal half sine wave pulse. Failure to meet the requirements of 3.6.14 shall constitute failure of this test.

4.6.14 Vibration (1), (Type I multiplier). Secure the multiplier to the vibration table. Vibrate the multiplier with simple harmonic motion in three mutually perpendicular planes at a constant frequency of 30 Hz with an amplitude of 0.125 inch total excursion for 5 minutes in each plane. There shall be no voltage applied to the multiplier during the vibration. After vibrating, operate the multiplier with an input of 2800V, ±15 volts (peak to peak). Failure to meet the requirements of 3.6.15 shall constitute failure of this test.

4.6.15 Vibration (2), (Type I and II multipliers). Secure the multiplier to the vibration table. Vibrate the multiplier with simple harmonic motion in three mutually perpendicular planes over a frequency range of 10 to 55 Hz with an amplitude of 0.10 inch total excursion for 10 minutes in each plane. After vibration, operate the multiplier with an input of 2800V, ±15 volts (peak to peak). Failure to meet the requirements of 3.6.16 shall constitute failure of this test.

4.6.16 Mean time between failure. Sample multipliers shall be selected at random for life testing each 30 days. The sample size shall be any quantity of multipliers accepted in the preceeding 30 days. Increasing the sample size will decrease the test time required to reach a decision as to lot acceptability. For first article life testing no replacements will be permitted; however, for life testing of production multipliers, replacement will be allowed and is desireable. Sample multipliers selected for life testing shall be selected from a lot which has been subjected to and met the requirements of 4.5.1 and 4.5.3 inspections. Life testing shall be conducted at room temperature (see 6.3) with the multipliers mounted in a grounded tube simulator . The multipliers shall be operated intermittently as follows:

a.	ON time	55 minutes
b.	OFF time	5 minutes

The total number of operating hours for all units, failed and unfailed, original and replacement, shall be monitored and recorded. Once each 24 hours the following inspections shall be performed.

NOTE I : During weekends and other periods of general cessation of plant activity, this time period may be extended not to exceed 96 hours.

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Test	Test paragraph
Continuity	4.6.3
Ground probe resistance	4.6.6
Input voltage waveform	4.6.8
D.C. output voltage	4.6.4
Multiplier input capacitance	4.6.1
Resistance (third stage, limiting)	4.6.5

Failure of a multiplier to meet the inspections above shall be considered a failure and the hours accumulated by the multiplier, to the last successful check, shall be used in the totalized hours for the lot under test. Test shall be accomplished in accordance with MIL-STD-781 test level A1, test plan VI. Each voltage multiplier which has failed shall be analyzed by the supplier for a cause of failure, including a radiograph of the multiplier prior to any disassembly. The Government shall be advised of all failures, causes, and steps taken by the supplier to correct the causes. Failure to meet the requirements of 3.6.10 shall constitute failure of this test.

4.7 Inspection of Packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Multiplier, High Voltage D.C. shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Multiplier, High Voltage D.C. shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Multiplier, High Voltage D.C.. Each multiplier shall be individually unit packed method III as follows: cushion each unit on all surfaces with polyurethane foam type I, class 2, conforming to MIL-P-26514. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

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5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Multiplier, High Voltage D.C. unit packed as specified in 5.1 shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inchesor more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Clsoure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.2 Level B. A quantity of Multiplier, High Voltage D.C., unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Commercial packing Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356

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6. NOTES

6.1 Intended use. The voltage multiplier is intended to be used to provide high voltage dc operating potential to the three single stage image intensifier modules which form the types MX-7854/UV and MX-8501/UV (25 MM) Image Intensifier Assembly, and the type MX-7856/UV (40 MM) Image Intensifier Assembly.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article tests are to be conducted.
c. Time frame required for submission of first article test reports.
d. Time frame required for approval of the first article test reports by the contracting officer.
e. Production delivery schedule.
f. Level A, level B or Commercial preservation and packing (see Section5).
g. Actions required relative to Group C failures.
h. When the rough handling test is required.
i. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.3 Definitions. For the purpose of this specification the following definitions apply.

6.3.1 Damage. Damage is defined as the following:

a. Electrical failure or malfunction, including arcing, flashing, or corona.
b. Cracks, voids, blisters, scratches or deformation.

6.3.2 No-load. Multiplier output leads terminated with silicone rubber corona shields.

6.3.3 "g". g is defined as an acceleration or deceleration of 32.17 feet per second per second.

6.3.4 Room temperature. Room temperature is defined as 23°C, ±3°C.

Custodian:
Army - EL
Navy - EC
Air Force: 99

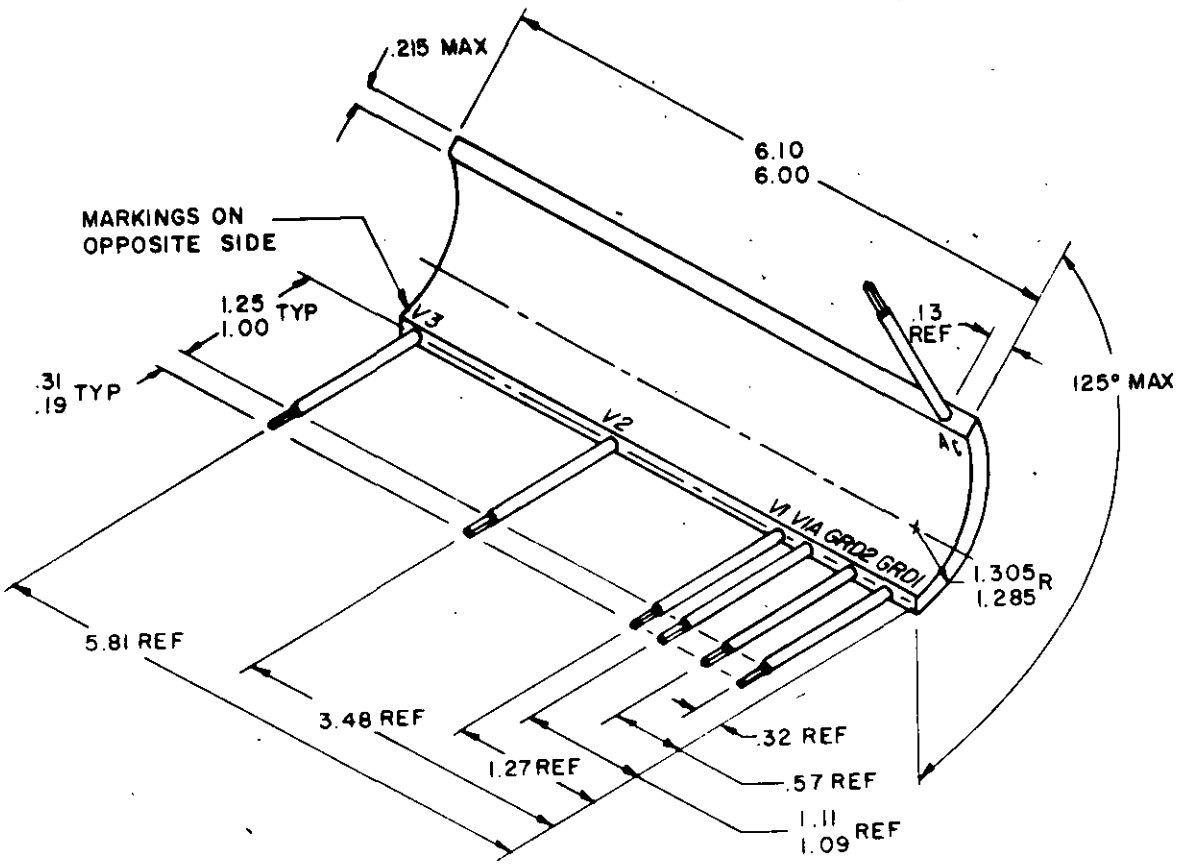
Preparing activity:
Army - EL

Review activity:
Army - MI

Project No. 5855-0019

User activity:
Army - AV

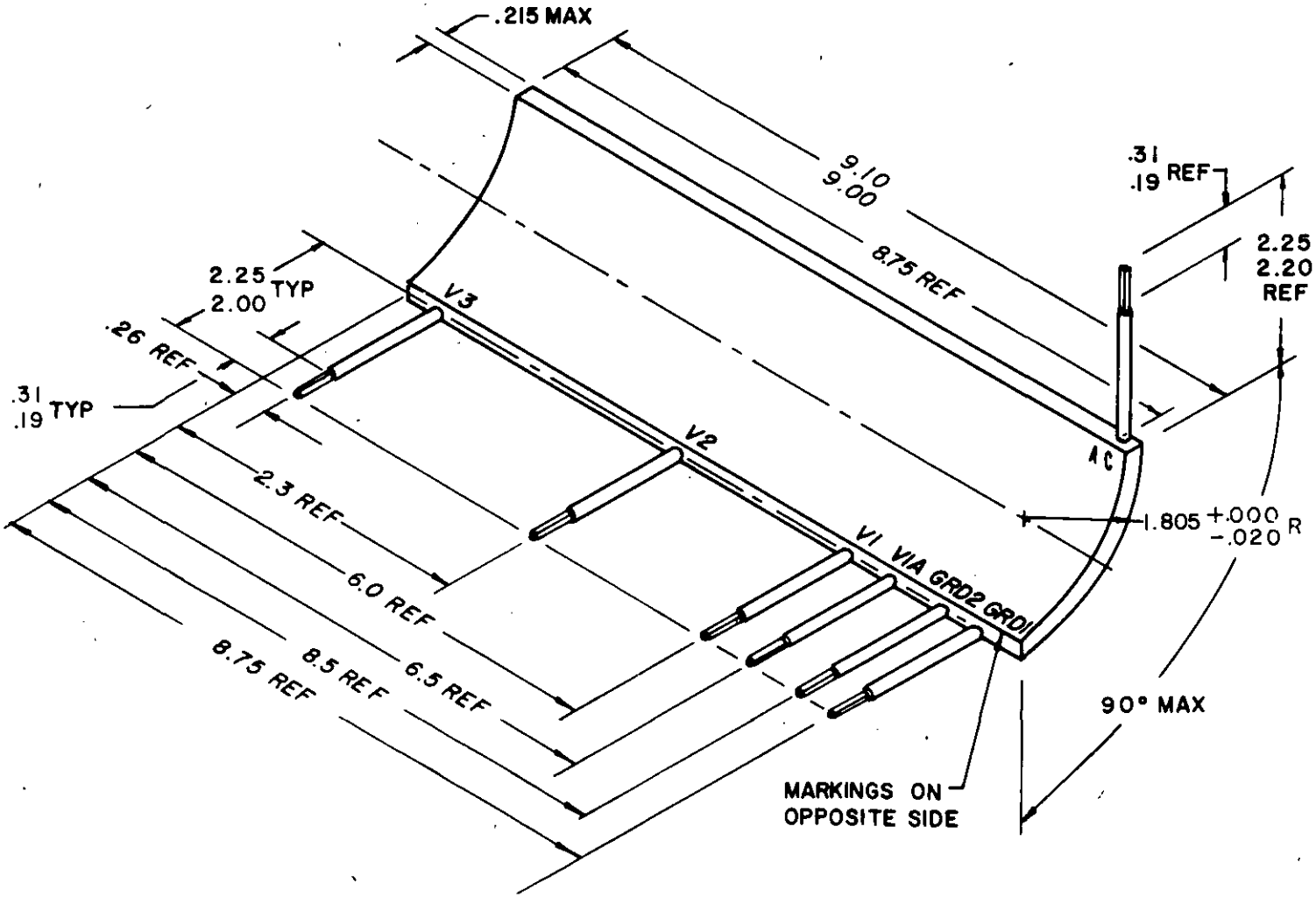
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TYPE I MULTIPLIER

FIGURE 1

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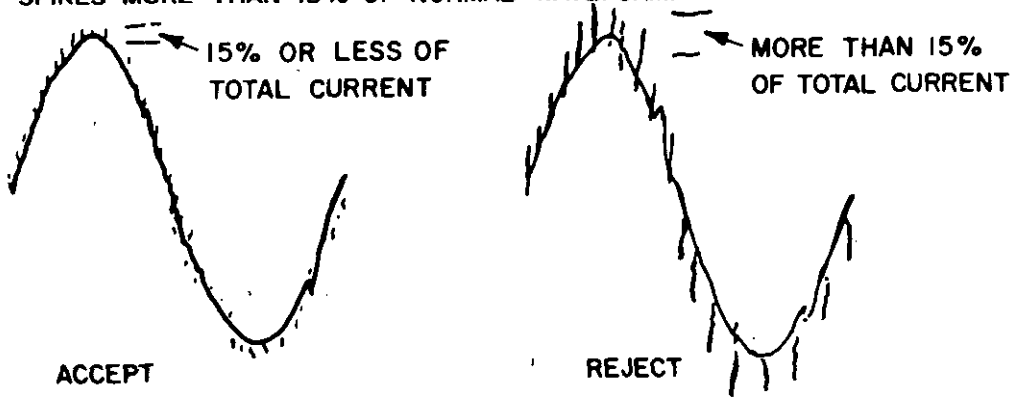
TYPE II MULTIPLIER
FIGURE 2

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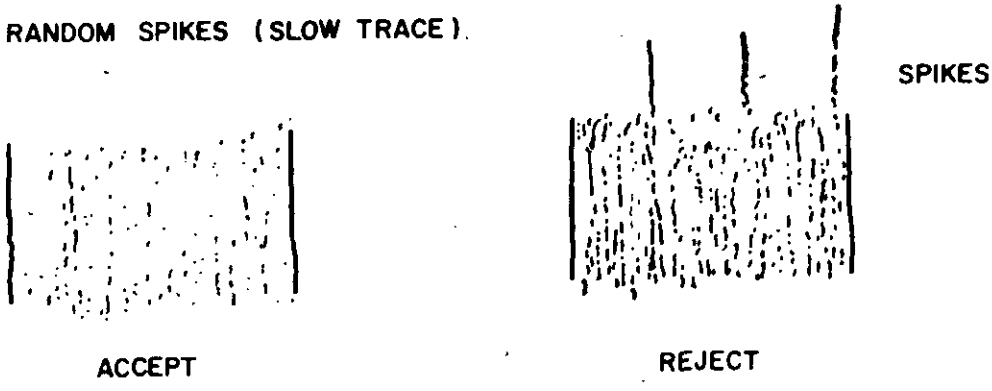
a. ABRUPTLY CHANGING CHARGING CURRENT



b. SPIKES MORE THAN 15% OF NORMAL WAVEFORM.



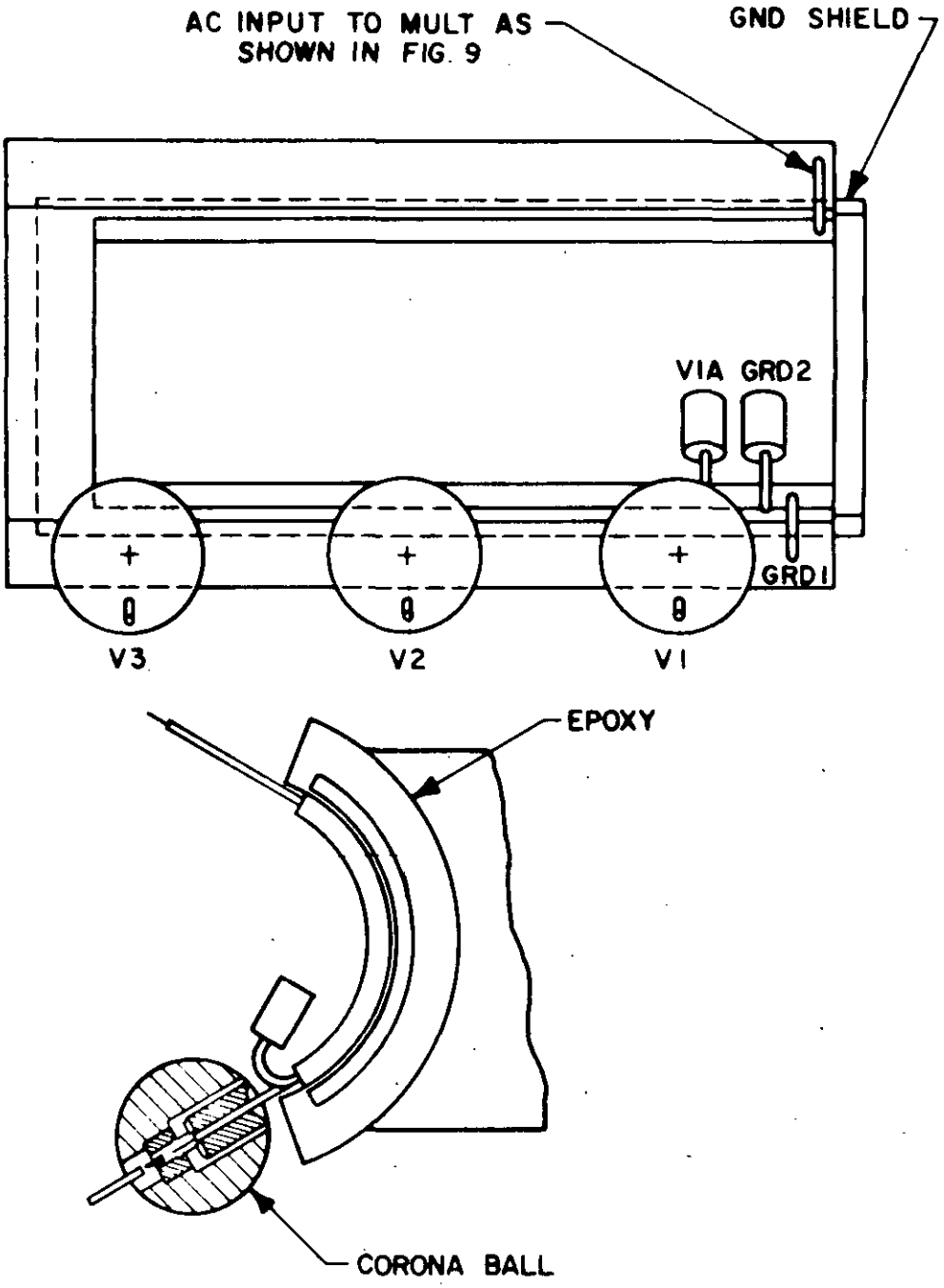
c. RANDOM SPIKES (SLOW TRACE).



NOTE:
SPIKES CAN BEST BE SEEN WITH THE INTENSITY TURNED UP AND THE CURRENT WAVEFORM NEAR THE BOTTOM OF THE SCREEN

FIGURE 3

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TUBE SIMULATOR WITH CORONA BALLS

FIGURE 4

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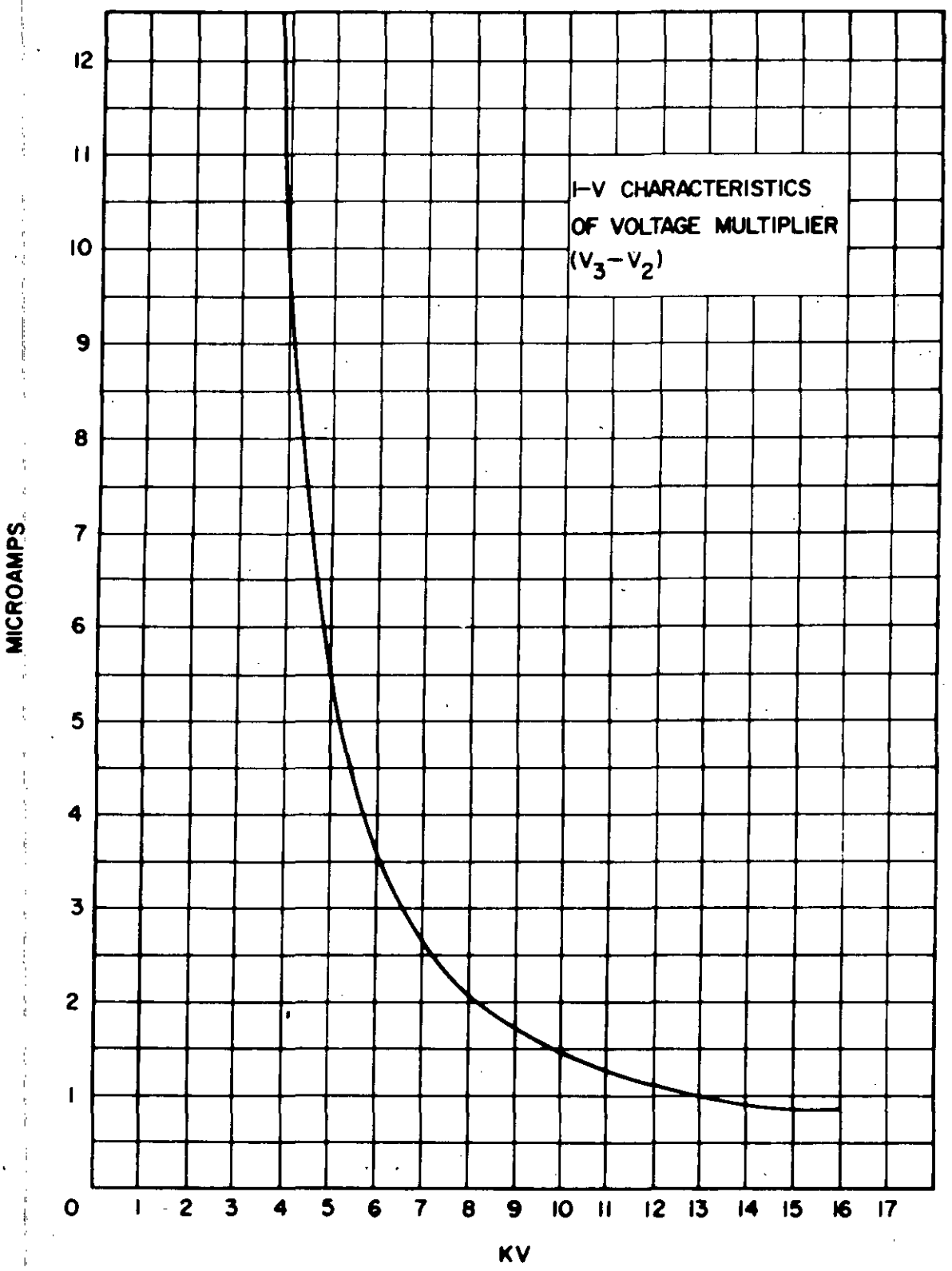


FIGURE 5

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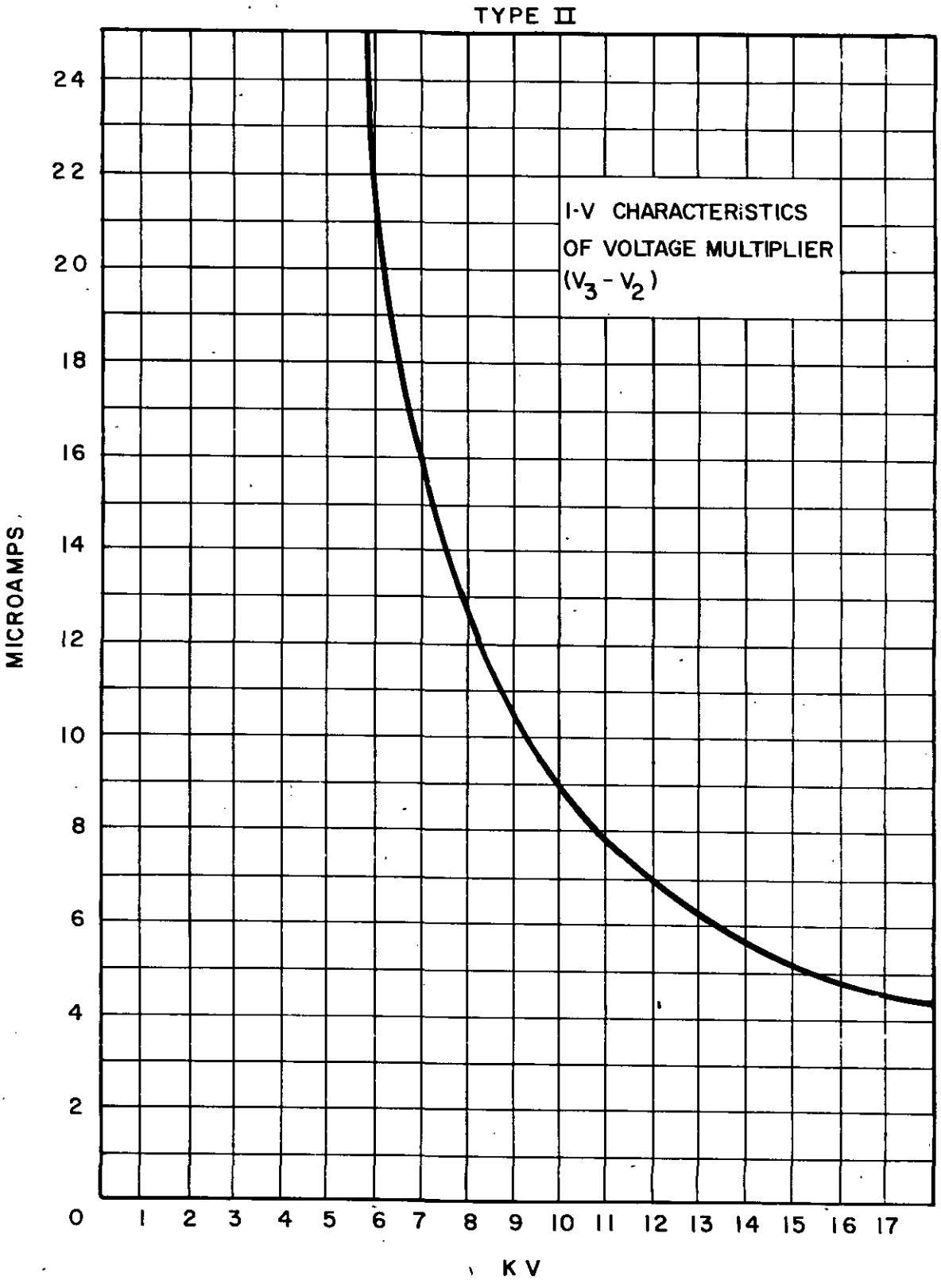
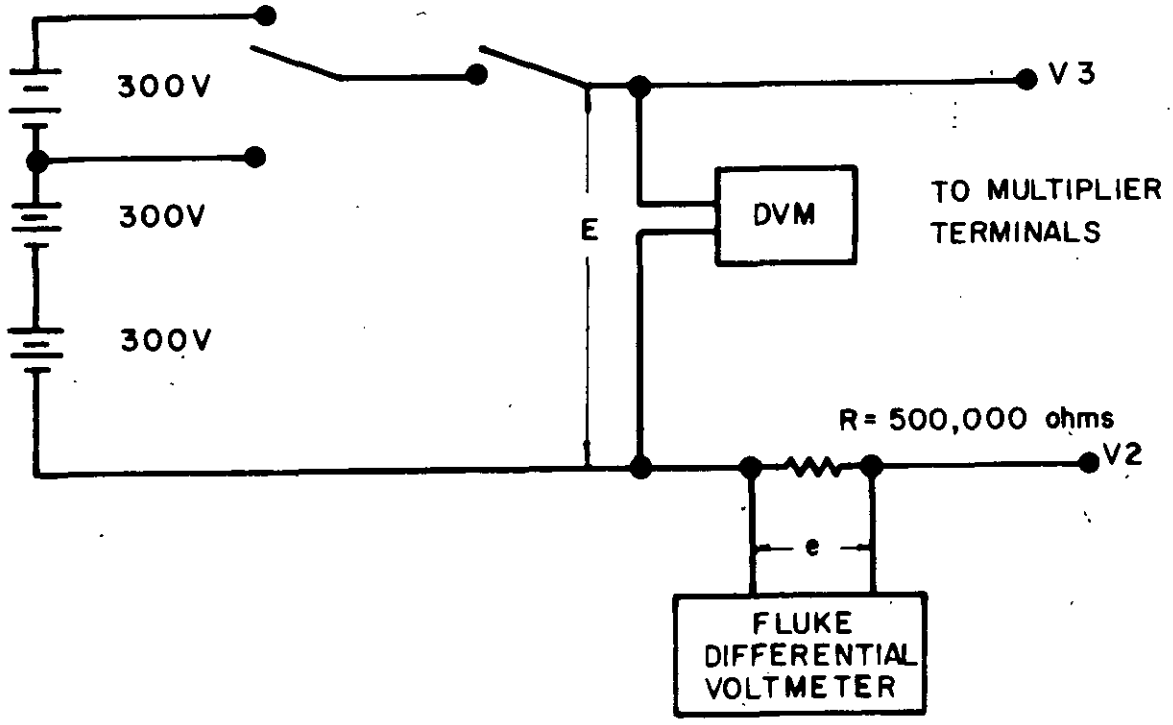


FIGURE 5a

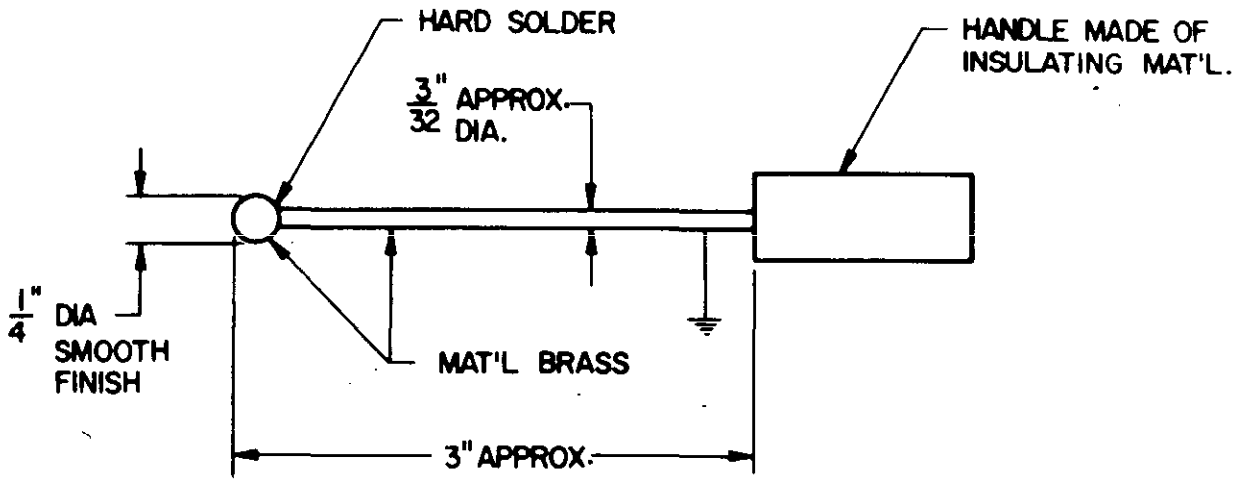
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$$R_{multiplier} = \frac{E900 - E600}{e900 - e600} \times 5 \times 10^5$$

TEST CIRCUIT
FIGURE 6

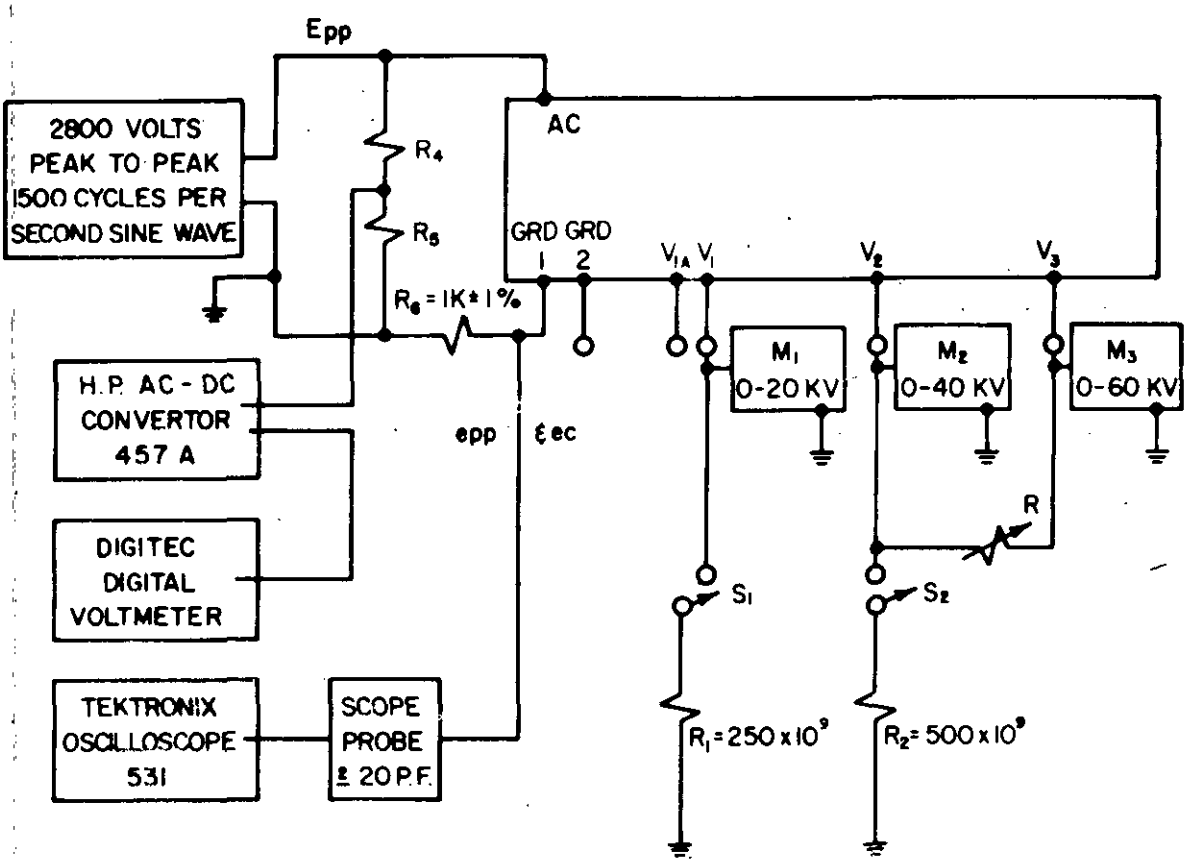
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GROUND PROBE

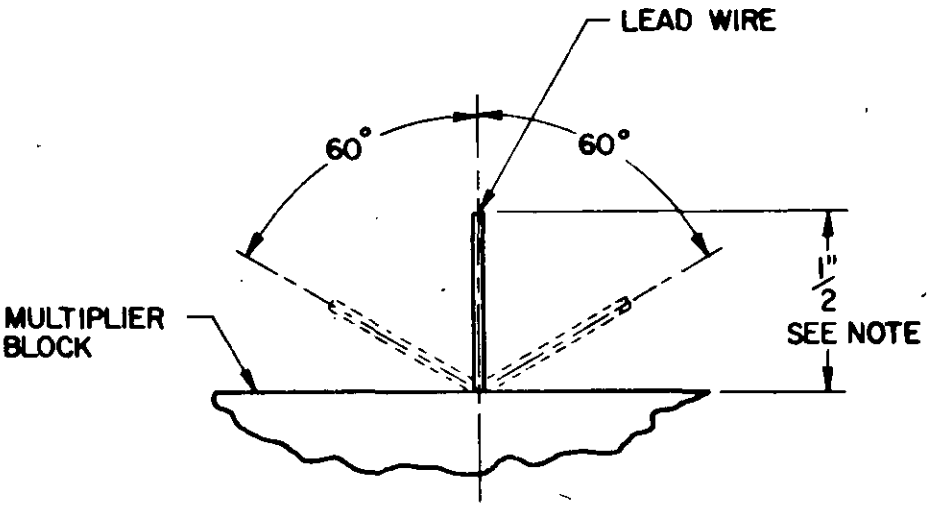
FIGURE 7

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TEST CIRCUIT
FIGURE 8

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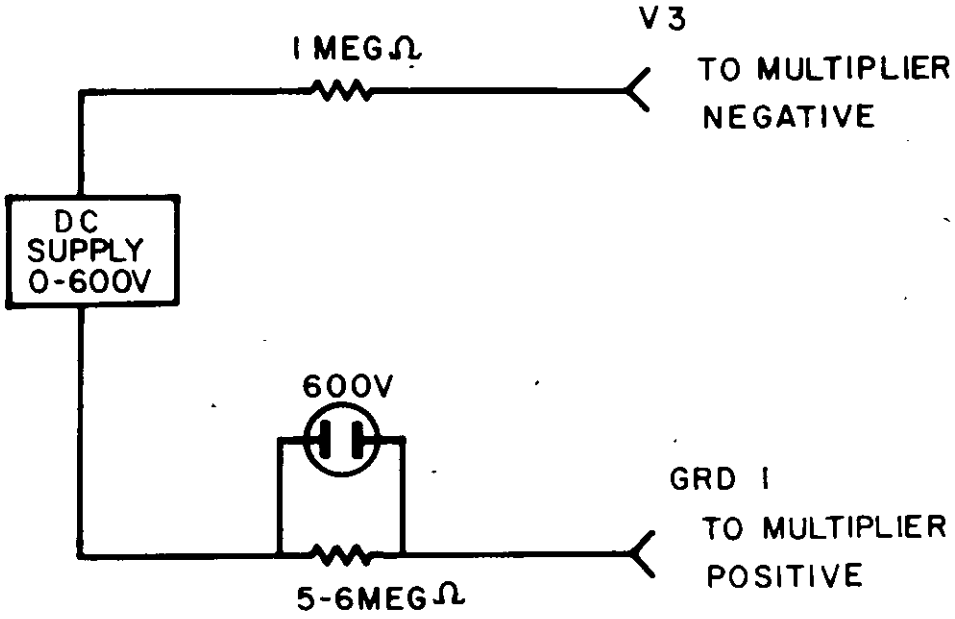


NOTE: 1/2" DIMENSION SPECIFIES POINT AT WHICH FORCE IS APPLIED TO BEND LEAD. LEAD IS ACTUALLY LONGER.

LEAD FLEXING TEST

FIGURE 9

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TEST CIRCUIT

FIGURE 10

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-55520A MULTIPLIER: HIGH VOLTAGE, DIRECT CURRENT

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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